GMI German-Malaysian Institute

DIPLOMA PROGRAMME

PRACTICAL TEST 2

Academic Period: July 2025

Program	:	**NETWORK SECURITY**
Course	:	**SCALING NETWORK**
Code	:	**NTC 1093**
Duration	:	**3 HOURS**
Total Marks	:	**100**

GENERAL INSTRUCTIONS:

1. Write down clearly (with capital letters) your name, trade, group and the date of examination at the bottom of this page.
2. Do not open this question booklet until you are told to do so by the invigilator.
3. Read the specific instructions and questions carefully before answering.

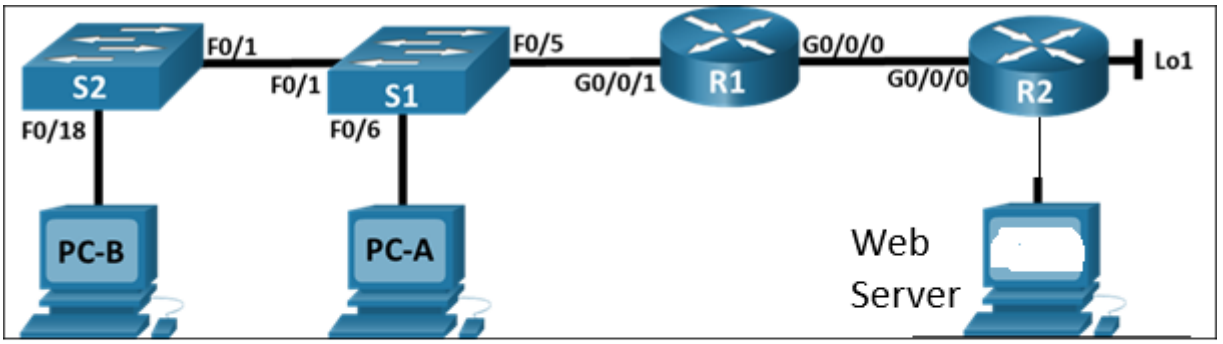
This paper consists of **7** printed pages including this page.

Student's Name	:	**ABDUL HARIZ BIN MOHD SARNUDIN**
	:	**MUHAMMAD DANIAL BIN NORHISAM**
	:	**PUTRI NABILA BINTI ZAINOL ABIDDIN**
Semester/Trade/Group	:	**3/DCBS/1**
Date	:	**4 NOVEMBER 2025**

No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Setup Network	2	4	P3	3			
2	Configuration	2	4	P3	3			
3.	Verification	2	4	P3	3			
4.	Interpretation	2	4	P3	3			
5	Knowledge	2	4	P3	3			
				Total	15			
				Total Marks	/15			
	Total		**PO 2**		/15			
				Signature				

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NO.	CRITERIA	APPROACHES EXPECTATIONS [1]	MEETS EXPECTATIONS [2]	EXCELLENT [3]	MARKS
PRACTICAL TEST (1)					
1	**Setup Network** Cabling and setup network	Unable to setup network without major assistance.	Able to setup network with little assistance.	Able to setup network without assistance.	
2	**Configurations:** Solving the lab configuration PAT and ACL	Unable to configure most of the PAT and ACL	Able to partially configure both PAT and ACL	Able to configure all PAT and ACL	
3	**Verification:** Test connection	Unable to verify most of the configurations.	Students able to verify most of the configurations.	Student able to verify all configurations.	
4	**Interpretation:** Interpreting findings.	Unable to interpret most of the findings and result.	Able to interpret most of the findings and result.	Able to interpret all the findings and result clearly and comprehensively.	
5	**Knowledge**	Only restatement of the instruction without commenting on the expected key point. Incorrect judgment/arguments are used.	Shows a little understanding of what the practical was about.	Shows a full understanding of what the practical was about.	
	TOTAL				



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	S0/0/0	209.165.201.230	255.255.255.0	
	G0/1	192.168.1.1	255.255.255.0	
R2 (ISP)	S0/0/0	209.165.201.231	255.255.255.0	
	G0/1	209.165.200.1	255.255.255.224	
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.1.3	255.255.255.0	192.168.1.1
Web Server	NIC	209.165.200.5	255.255.255.224	209.165.200.1

Part 1: Build the Network and Configure Basic Device Settings

Step 1: Cable the network as shown in the topology.

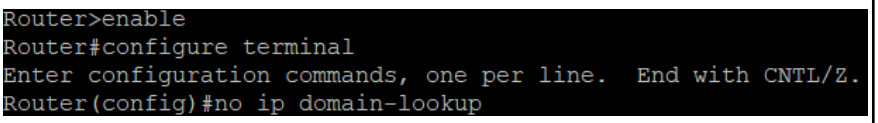
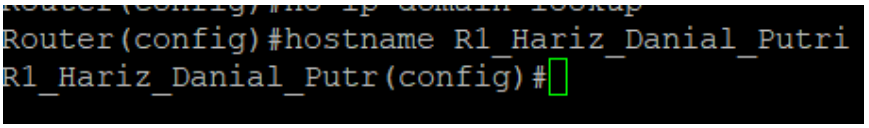
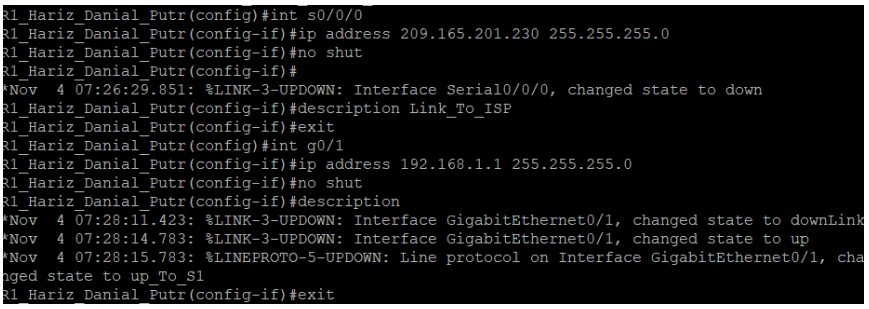
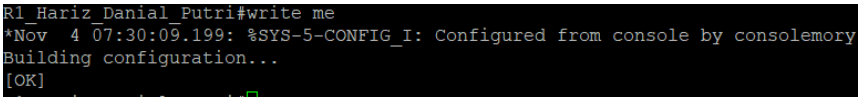
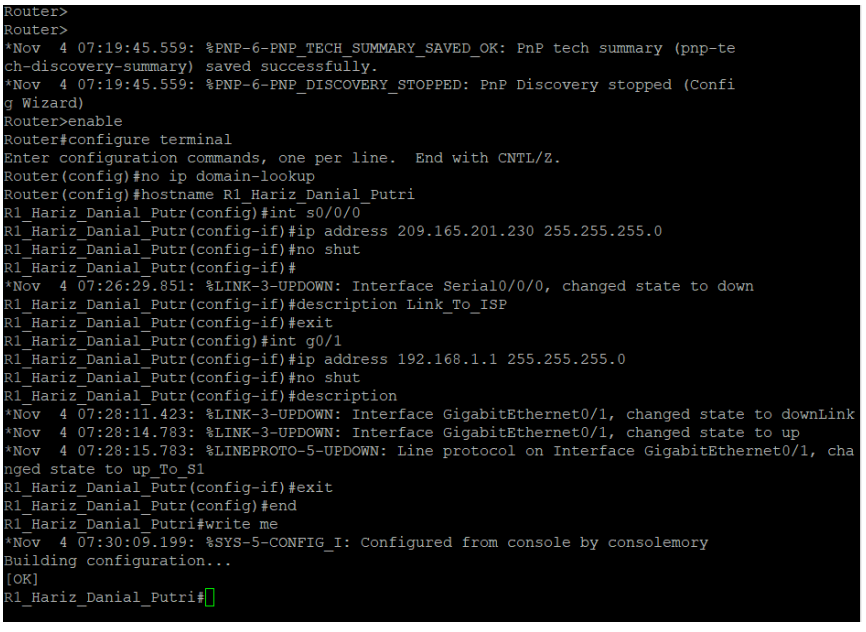
Step 2: Configure PC hosts.

Step 3: Initialize and reload the routers and switches as necessary.

Step 4: Configure basic settings for each router.

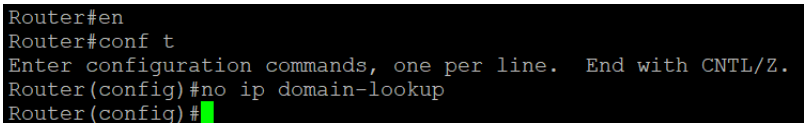
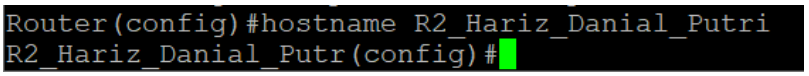
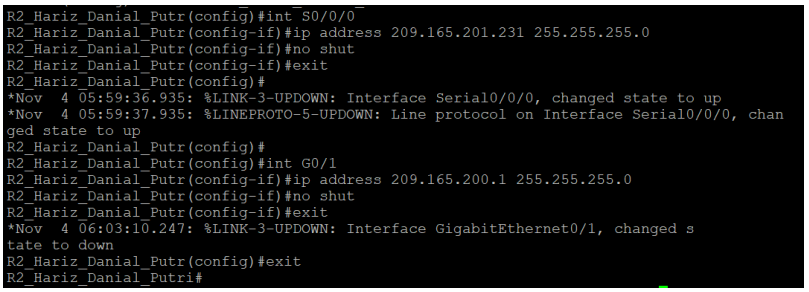
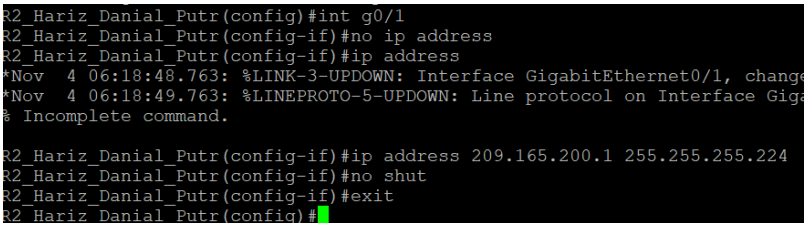
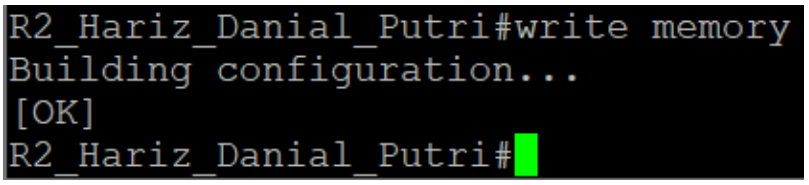
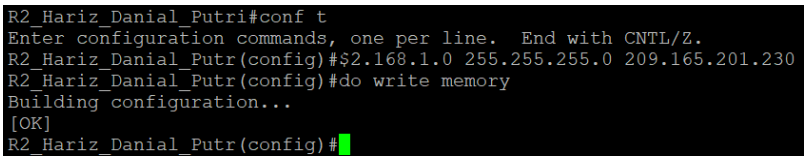
Step 5: DO NOT ERASE CONFIGURATION BEFORE TTO CHECKS YOUR ANSWER

Configuration tasks for router **R1** include the following:

Configuration Item or Task	Specification
1.Disable DNS lookup	
2.Router name	
3.IP addressing on each interface	
4.Save the configuration	
5.Screen capture of all above tasks	

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Configuration tasks for router **R2** include the following:

Configuration Item or Task	Specification
1.Disable DNS lookup	
2.Router name	
3.IP addressing on each interface	 Fix g0/1 : 
4.Save the configuration	
5.Configure default route from R1 to R2	

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6.Screen capture of all above tasks

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip domain-lookup
Router(config)#hostname R2_Hariz_Danial_Putri
R2_Hariz_Danial_Putr(config)#int S0/0/0
R2_Hariz_Danial_Putr(config-if)#ip address 209.165.201.231 255.255.255.0
R2_Hariz_Danial_Putr(config-if)#no shut
R2_Hariz_Danial_Putr(config-if)#exit
R2_Hariz_Danial_Putr(config)#
*Nov  4 05:59:36.935: %LINK-3-UPDOWN: Interface Serial0/0/0, changed state to up
*Nov  4 05:59:37.935: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, chan
ged state to up
R2_Hariz_Danial_Putr(config)#
R2_Hariz_Danial_Putr(config)#int G0/1
R2_Hariz_Danial_Putr(config-if)#ip address 209.165.200.1 255.255.255.0
R2_Hariz_Danial_Putr(config-if)#no shut
R2_Hariz_Danial_Putr(config-if)#exit
*Nov  4 06:03:10.247: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed s
tate to down
R2_Hariz_Danial_Putr(config)#exit
R2_Hariz_Danial_Putri#
*Nov  4 06:03:17.419: %SYS-5-CONFIG_I: Configured from console by console
R2_Hariz_Danial_Putri#
R2_Hariz_Danial_Putri#write memory
Building configuration...
[OK]
R2_Hariz_Danial_Putri#ip route 192.168.1.0 255.
255.255.0 209.165.201.230
                         ^
% Invalid input detected at '^' marker.

R2_Hariz_Danial_Putri#ip route 192.168.1.0 255.
255.255.0 209.165.201.230
                         ^
% Invalid input detected at '^' marker.

R2_Hariz_Danial_Putri#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2_Hariz_Danial_Putr(config)#$2.168.1.0 255.255.255.0 209.165.201.230
R2_Hariz_Danial_Putr(config)#do write memory
Building configuration...
```

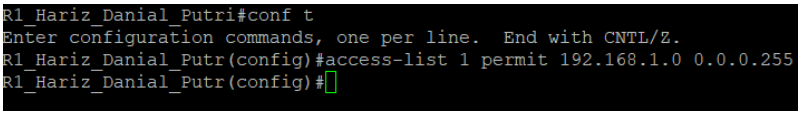
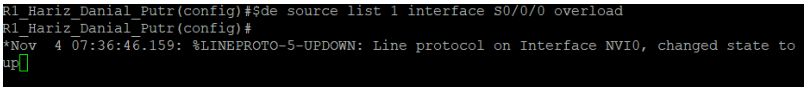
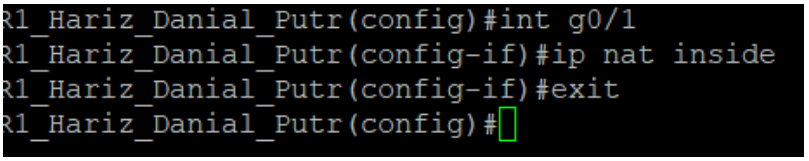
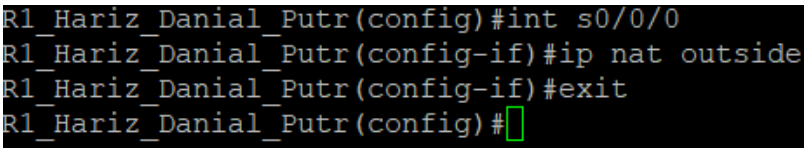
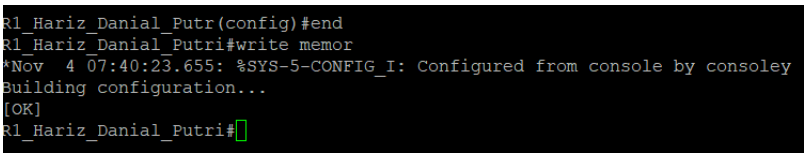
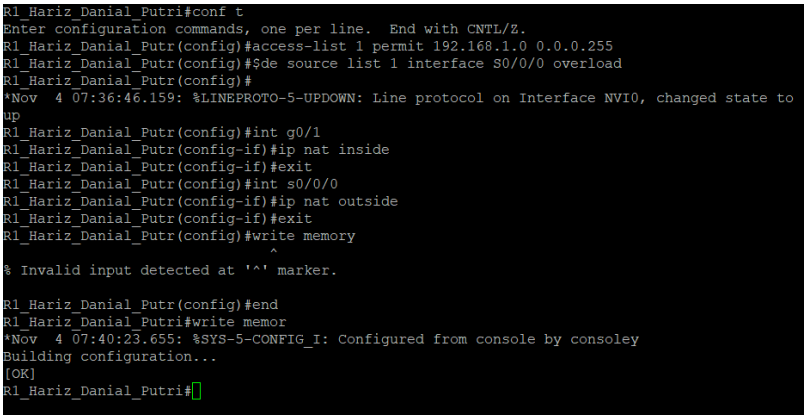
```
Building configuration...
[OK]
R2_Hariz_Danial_Putr(config)#
```


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Part 2: Configure and verify PAT (NAT Overload) for IPv4

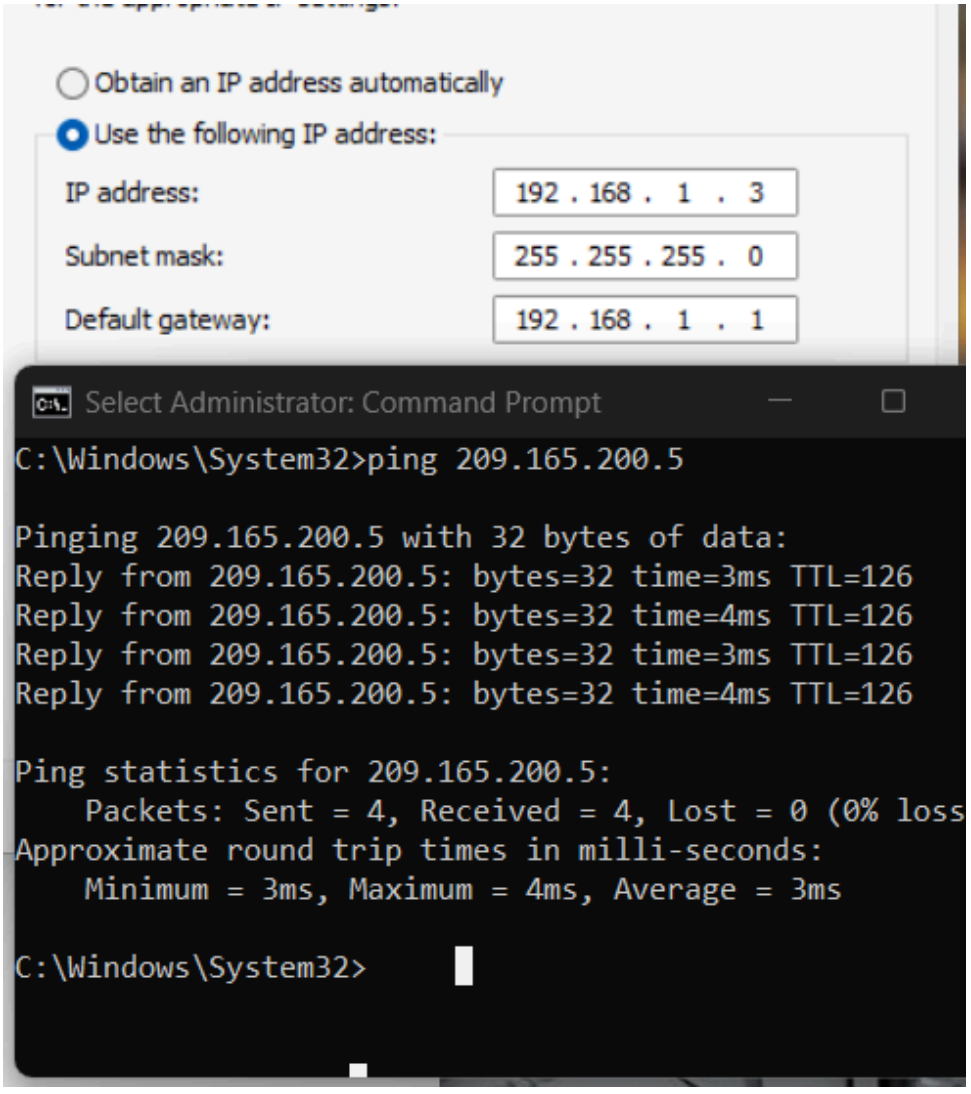
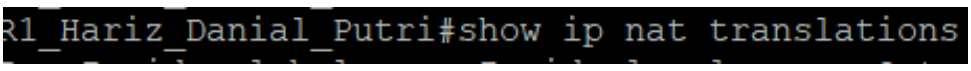
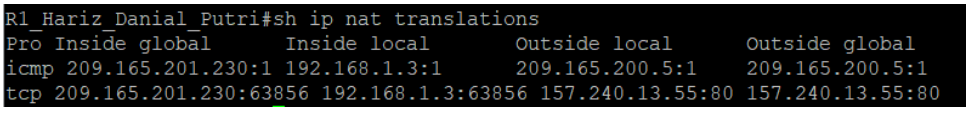
Step 1: Configure NAT Overload on R1.

Configuration and verify NAT Overload for IPv4

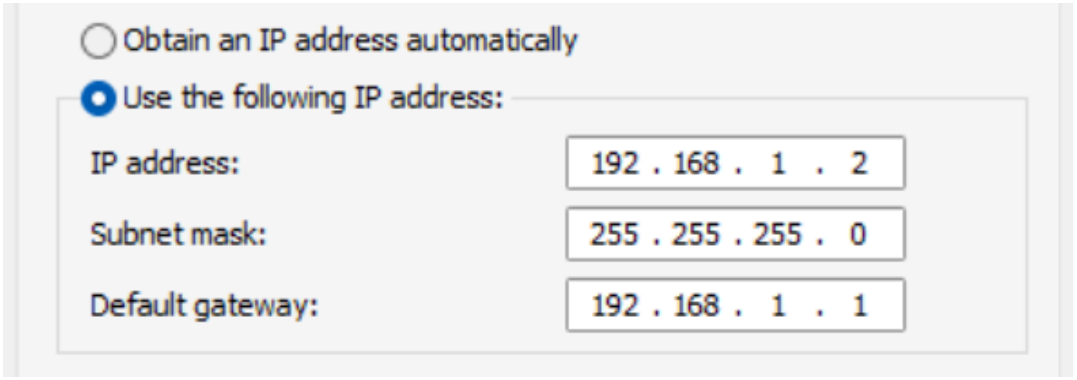
Configuration Item or Task	Specification
1.Configure ACL to ensure all devices on the R1 LAN eligible for translation	
2.Create the NAT overload using interface R1's S0/0/0	
3.Define the inside interface.	
4.Define the outside interface.	
5.Save the configuration	
6. Screen capture of all above tasks	

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Step 6: Test and verify the configuration

Configuration Item or Task	Specification
1.Ping Web Server from PC-B	Choose 1: Successful/ Unsuccessful
2.Screen capture the result	
3.Write command to display the NAT table on R1	
4.Issue the above command with result	
5.What type of NAT address is the translated address?	The IP address 209.165.201.230 is the public IP address that the NAT device substitutes for the original private Inside Local address 192.168.1.3 when traffic leaves the internal network.
6.What was the inside local address of PC-B translated to?	For outgoing traffic, the Inside Local address of the host (192.168.1.3) was mapped to the Inside Global address 209.165.201.230 using port number 63856

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Configuration Item or Task	Specification
1.Ping Web Server from PC-A	Choose 1: Successful/ Unsuccessful
2.Screen capture the result	 C:\Users\Lenovo>ping 209.165.200.5 Pinging 209.165.200.5 with 32 bytes of data: Reply from 209.165.200.5: bytes=32 time=3ms TTL=126 Reply from 209.165.200.5: bytes=32 time=3ms TTL=126 Reply from 209.165.200.5: bytes=32 time=4ms TTL=126 Reply from 209.165.200.5: bytes=32 time=3ms TTL=126 Ping statistics for 209.165.200.5: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 3ms, Maximum = 4ms, Average = 3ms
3.Write command to display the NAT table on R1	Command is show ip nat translations The show ip nat translation command is used on Cisco routers to verify and troubleshoot Network Address Translation (NAT). It displays the router's active NAT table, which is a real-time record of every ongoing conversation being translated. The output shows the mapping between the private IP addresses and ports (Inside Local) and the public IP addresses and ports (Inside Global) they are translated to. This command is crucial for confirming that both static and dynamic NAT/PAT (Port Address Translation) are working correctly for traffic passing through the router.
4. Issue the above command with result	R1_Hariz_Danial_Putri#sh ip nat translations Pro Inside global Inside local Outside local Outside global icmp 209.165.201.230:1 192.168.1.3:1 209.165.200.5:1 209.165.200.5:1 tcp 209.165.201.230:63856 192.168.1.3:63856 157.240.13.55:80 157.240.13.55:80

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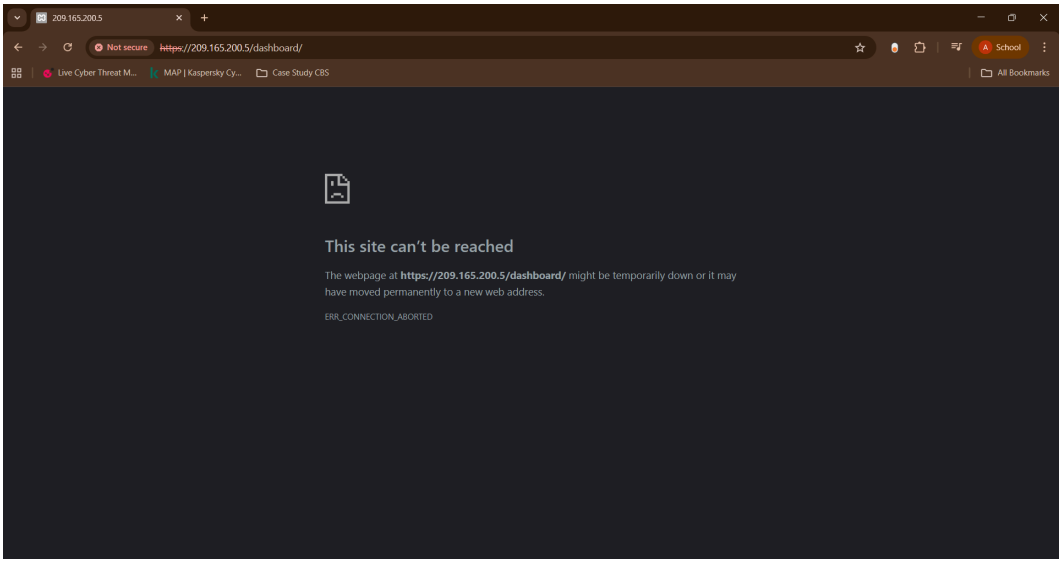
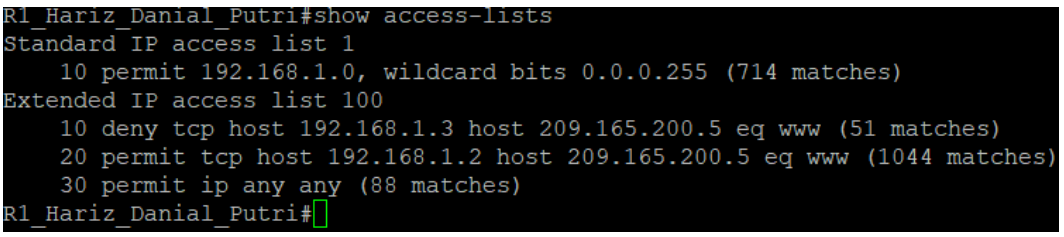
Part 3: Configure and verify ACL

Step 1: Apply ACL for R1 only

Configuration Item or Task	Specification
1.Ensure PC A can access Web Server	R1_Hariz_Danial_Putr(config)#$t tcp host 192.168.1.2 host 209.165.200.5 eq 80 R1_Hariz_Danial_Putr(config)#
2.Ensure PC B cannot access Web Server	R1_Hariz_Danial_Putr(config)#$tcp host 192.168.1.3 host 209.165.200.5 eq 80 R1_Hariz_Danial_Putr(config)#
3.All other traffic is allowed	R1_Hariz_Danial_Putr(config)#access-list 100 permit ip any any R1_Hariz_Danial_Putr(config)#

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Step 2: Test and verify the configuration

Configuration Item or Task	Specification
1.Browse website from PC A	
2.Browse website from PC B	
3.Write command to verify ACL (should have matches)	

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Part 4: Cleanup

Before turning off power to the routers, remove the NVRAM configuration files (if saved) from all devices.

Disconnect and neatly put away all cables that were used in the final.

PLEASE SWITCH OFF & CLEANUP ALL DEVICES.

CABLES TO BE NEARTLY ARRANGED.

FAIL TO DO SO WILL AFFECT YOUR MARK.

Submission in VLU:

1) Lab sheet (in **pdf format**)
2) R1 (Abdul Hariz bin Mohd Sarnudin)
3) R2 (Putri Nabila binti Zainol Abiddin)
4) Network setup (Muhamamd Danial bin Norhisam)

-THE END-

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